Resistors

A resistor is a passive electrical component with the primary function to limit the flow of electric current. It works by impeding the flow of electrons, converting electrical energy into heat, and is used for purposes like current control, voltage division, and protecting sensitive components. The unit of measurement for resistance is the ohm (**Ω)**



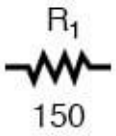
Schematic Symbol

In electronic circuit diagrams (schematics), **resistors** are components that limit or control the flow of electric current. Their **values** are given in **ohms (Ω)**, for example, 100 Ω, 4.7 kΩ, or 1 MΩ. These numbers are usually written **next to** the resistor symbol to indicate its resistance value.

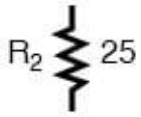
If a circuit has **multiple resistors**, each one is assigned a unique label (called a reference designator) for easy identification, such as:

- **R1** = first resistor
- **R2** = second resistor
- **R3** = third resistor
  …and so on.

In circuit diagrams, resistors can appear **either horizontally or vertically**

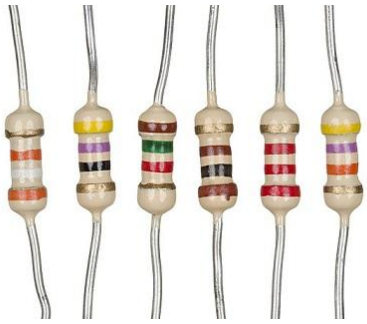


This is resistor "R_1" with a resistance value of 150 ohms.



This is resistor "R_2" with a resistance value of 25 ohms.

Resistors look very different from the **zig-zag** or **rectangular symbols** used in circuit diagrams. The schematic symbol is just a convenient way to represent a resistor on paper, but actual resistors come in a variety of **shapes, sizes, and materials**, depending on their purpose and power rating.



Types of Resistors

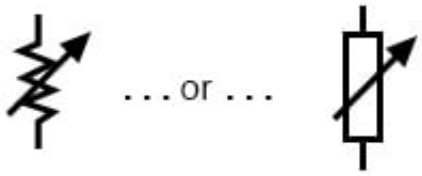
- Fixed Resistors

  These resistors have a **set resistance value** that cannot be adjusted. They are the most commonly used type.



- Variable Resistors

These resistors can change their resistance manually or automatically.



a. Potentiometer

- ❖ Has a rotating knob or slider to vary resistance.
- ❖ Works as a **voltage divider**.
- ❖ **Used in:** Volume controls, dimmers, and tuners.

b. Rheostat

- ❖ Similar to a potentiometer but used to **control current**.
- ❖ Usually has two terminals.
- ❖ **Used in:** Motor speed control, lamps, or heaters.

c. Trimmer (Preset Resistor)

- ❖ A small adjustable resistor set once during calibration.
- ❖ **Used in:** Fine-tuning circuits during manufacturing.

Resistor Color Code

Most resistors are so small that writing their values in numbers isn't possible, so they're marked with **colored bands** to indicate their resistance value and tolerance. This is known as the **resistor color code**.

Each color represents a number, multiplier, or tolerance.

Color	Color	1st Band	2nd Band	3rd Band Multiplier	4th Band Tolerance
Black		0	0	x1Ω	
Brown		1	1	x10Ω	±1%
Red		2	2	x100Ω	±2%
Orange		3	3	x1kΩ	
Yellow		4	4	x10kΩ	
Green		5	5	x100kΩ	±0.5%
Blue		6	6	x1MΩ	±0.25%
Violet		7	7	x10MΩ	±0.10%
Grey		8	8	x100MΩ	±0.05%
White		9	9	x1GΩ	
Gold				x0.1Ω	±5%
Silver				x0.01Ω	±10%

How to Read a 4-Band Resistor

A 4-band resistor has:

1. 1st band → 1st digit
2. 2nd band → 2nd digit
3. 3rd band → multiplier
4. 4th band → tolerance

Example:

- Bands: **Red (2), Violet (7), Orange (×1,000), Gold (±5%)**
- Calculation: 27 × 1,000 = 27,000Ω = 27 kΩ ±5%
  ✅ **Answer:** 27 kΩ ±5%

🧩 **5-Band Resistor (for Precision Resistors)**

Used for more accurate (low-tolerance) resistors.

1. 1st band → 1st digit

2. 2nd band → 2nd digit
3. 3rd band → 3rd digit
4. 4th band → multiplier
5. 5th band → tolerance

Example:
Bands: **Brown (1), Black (0), Black (0), Red (×100), Brown (±1%)**
Calculation: 100 × 100 = 10,000Ω = 10 kΩ ±1%

Tips for Reading:

- Always start reading from the end **closer to the first color bands** (the tolerance band is usually spaced farther apart).
- Use a **multimeter** to double-check your readings.

References:

https://www.allaboutcircuits.com/textbook/direct-current/chpt-2/resistors/

https://www.theengineeringknowledge.com/wp-content/uploads/2025/09/Resistors-Color-Code-Chart.jpg